

Module 1

Introduction to Embedded Systems and Microcontrollers

Your Instructor

Professor Nilufar Poshtiban, PhD

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How to contact your instructor?

- Send a message on Blackboard.
- **Do not send** an email to Nilufar.Poshtiban@humber.ca for course-related inquiries.
- Office hours are by appointment only.

Class Delivery Mode

The format of this course in Winter 2026 is **in-person**:

- 2h **lecture**
- 2h **in-person lab**

Resources

- Textbook:

References	
Textbook 1	Mazidi, M., Naimi, S., Naimi, S., Chen, S. (2016). ARM Assembly Language Programming & Architecture (Volume 1) Second Edition. MicroDigitalEd. ISBN-13 978-0997925906
Textbook 2	Mazidi, M., Chen, S, Naimi, S., Naimi, S. (2017) TI TIVA ARM Programming for Embedded Systems, MicroDigitalEd. ISBN-13 978-0997925920

- Class notes and topical references will be provided on blackboard.
- Being an active learner in the classroom and lab is critical to success in this course.

Evaluation Plan

Summary	Weight
Midterm Exam	20%
Final Exam	30%
Quizzes (x2)	5%
Laboratory Assignments (x5)	30%
Bus Project	15%
Total	100%

A passing grade for this course is 60%.

Labs/Projects and Grades

- There are 5 lab assignments and 4 project reports.
- There may be a “pre lab” assignments that must be completed prior to attending the in-person lab.
- Hardware and code checkouts are required for labs. Submissions without the “Instructor Signature” will be only evaluated for the theory portion of the assignment. All checkouts will be done during lab hours.
- Written reports are required for all project reports.
- Partial credits will be given for partially finished projects and labs.
- No late submission is allowed, and the permission may be granted for right reasons.

WHERE ARE WE (IN THE PROGRAM MAP)?

YEAR 1	Term 1	CALC 1100 Calculus 1	ALGB 1100 Linear Algebra	ENGI 1000 Physics 1	ENGI 1010 Introduction to Engineering	ENGI 1021 Engineering in Society	ENGI 1021 Technical Communication and Critical Thinking for Engineers		
	22 Cr. / 27 Hrs.	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 6	Cr. 3 Hrs. 3	Cr. 3 Hrs. 3		
	Term 2	CALC 1500 Calculus 2 PR: CALC 1100	ENGI 1510 Engineering Design	ENGI 1500 Physics 2 PR: CALC 1100	ENGI 1511 Engineering Materials	ENGI 1515 Introduction to Programming	ENGI 1512 Teamwork and Leadership for Engineers		
	22 Cr. / 27 Hrs.	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 3 Hrs. 4	Cr. 4 Hrs. 5	Cr. 3 Hrs. 3		
YEAR 2	Term 3	DIFF 2100 Differential Equations PR: CALC 1500 ALGB 1100	MENG 2010 Applied Mechanics PR: ENGI 1000	MENG 2000 Electric Circuits Analysis	MENG 2020 Computer Programming PR: ENGI 1515	MENG 2030 Digital Electronics PR: ENGI 1500	ENGI 2000 Engineering Project Management		
	22 Cr. / 26 Hrs.	Cr. 3 Hrs. 3	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 3 Hrs. 3		
	Term 4	NUMM 2500 Numerical Methods PR: DIFF 2100	MENG 2500 Mechatronics Project CR: Microcontrollers	MENG 2510 Instrumentation & Measurement PR: MENG 2000	MENG 2520 Pneumatics & Hydraulics	MENG 2530 Microcontrollers PR: MENG 2000 MENG 2030	Degree Breadth		
	18 Cr. / 24 Hrs.	Cr. 4 Hrs. 5	Cr. 2 Hrs. 4	Cr. 3 Hrs. 4	Cr. 3 Hrs. 4	Cr. 3 Hrs. 4	Cr. 3 Hrs. 3		
YEAR 3	Term 5	MENG 3000 Statistics and Quality Assurance	MENG 3010 Electric Motors PR: ENGI 1500 MENG 2000	MENG 3020 System Modeling and Simulation PR: DIFF 2100	MENG 3065 Introduction to Artificial Intelligence (shared) PR: MENG 2510	MENG 3030 Robotics (Embedded Systems) PR: MENG 2530	MENG 3050 Power Transmission Components (Robotics Stream) PR: MENG 2010	ENGI 3000 Engineering Economics and Entrepreneurship	WORK 3590 Work Preparation
	21 Cr./25 Hrs. + 14 hrs Work Prep	Cr. 3 Hrs. 3	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 3 Hrs. 4	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 3 Hrs. 3	Cr. 0 Hrs. 14
	Term 6	MENG 3500 Programmable Logic Controllers PR: MENG 2020 MENG 2030	MENG 3510 Control Systems PR: DIFF 2100 MENG 3020	MENG 3520 Signal Processing PR: CALC 1500 MENG 2030	MENG 3530 Autonomous Vehicles (Embedded Systems) PR: MENG 3065	MENG 3540 Parallel Programming (Embedded Systems) PR: MENG 2020, MENG2530	MENG 3550 Robotics Electrical Systems (Robotics Stream) PR: MENG 2000	MENG 3560 Kinematics and Dynamics of Robots (Robotics Stream) PR: MENG 2010	ENGI 3500 Law, Ethics and Professional Practice
	23Cr. / 28 Hrs.	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 3 Hrs. 3	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 3 Hrs. 3	Cr. 3 Hrs. 3
12-MONTH Work Placements (Total Credits = 33) ENGI 3591, ENGI 3592, ENGI 3593 (11 credits and 420 hours each) Pre-req completion of WORK 3590 and be in Good standing									
YEAR 4	Term 7	ENGI 4000 Capstone Project 1 PR: 4 th year of study and 12-month co-op complete	MENG 4000 Machine Vision PR: MENG 3520	MENG 4010 Industrial Networking PR: MENG 3500	MENG 4040 Machine Learning (shared) (Embedded Systems) PR: MENG 3065	MENG 4030 Embedded Parallel Computing (Embedded Systems) PR: Parallel Programming	MENG 4060 Robot Embedded Programming (Robotics Stream) PR: MENG 3540, MENG 3550.	MENG 4050 Machine Condition Monitoring (Robotics Stream) PR: MENG 3050	Degree Breadth (Natural Science)
	23 Cr. / 29 Hrs.	Cr. 4 Hrs. 6	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 4 Hrs. 5	Cr. 3 Hrs. 3
	Term 8	ENGI 4500 Capstone Project 2 PR: Capstone Project 1	MENG 4530 User Experience and Multimedia (Embedded Systems) PR: ENGI 4010	MENG 4540 Internet of Things (Embedded Systems)	MENG 4550 Advanced Manufacturing & Automation (Robotics Stream) PR: MENG 3500, MENG 4010	MENG 4560 Robotic Cell Integration (Robotics Stream) PR: MENG 3500, MENG 4060, MENG 4010	Degree Breadth (Engineering Elective)		
	14 Cr. / 17 Hrs.	Cr. 4 Hrs. 6	Cr. 4 Hrs. 5	Cr. 4 Hrs. 3+2	Cr. 4 Hrs. 5	Cr. 4 Hrs. 3+2	Cr. 3 Hrs. 3		
Total Credits: 165 Hrs: 2842 (203 hrs *14wks) + WORK 3590 (0 credits, 14 hrs) and Works Terms (33 credits, 1260 hrs)									

Outline

- Part 1: Introduction to Embedded Systems
- Part 2: Introduction to Microcontrollers
- Part 3: Applications of Microcontrollers in Mechatronics
- Part 4: Q & A

What will you learn in this microcontroller course?

Main Points:

1. Understanding Microcontrollers

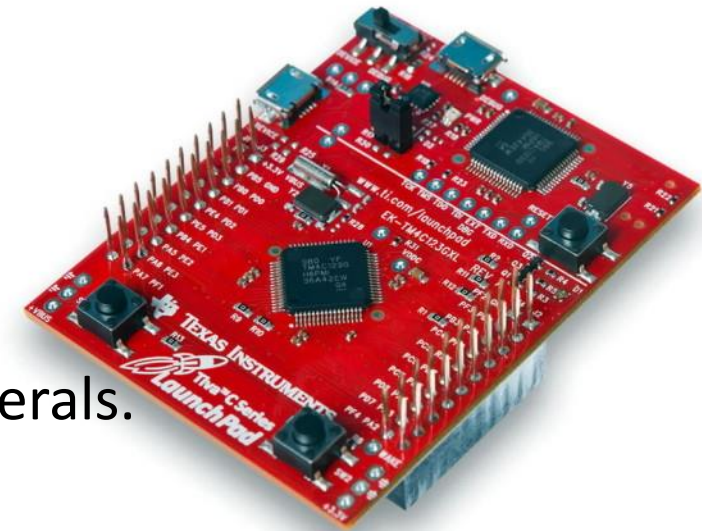
1. Core components and architecture.
2. Role in embedded systems and real-world applications.

2. Programming Microcontrollers

1. Using C and C++ for efficient development.
2. Introduction to assembly for low-level control.

3. Hardware Interfacing

1. Interacting with sensors, actuators, and peripherals.
2. Basics of analog and digital signal handling.



What will you learn in this microcontroller course?

4. Practical Applications

4. Hands-on projects to reinforce learning.
5. Real-world case studies in electromechanical systems.

5. Debugging and Optimization

1. Tools and techniques for troubleshooting.
2. Writing efficient and reliable embedded code.

6. Foundations for Advanced Learning

1. Building blocks for robotics, IoT, and mechatronics systems.

What is an embedded system?

Definition: An embedded system is a smaller computer system designed to perform a specific function within a larger system.

Key Characteristics:

- Real-time operation.
- Compact size and integration.
- Dedicated functionality.

Examples of embedded systems

- **Daily Life:**

- Washing machines, microwaves, and smart TVs.

- **Medical devices:**

- pacemakers, glucose meters.

- **Industrial:**

- CNC machines, conveyor belt, robotic arms, drones.

- **Automotive:**

- Engine Control Units (ECUs), ABS systems.

Components of Embedded Systems

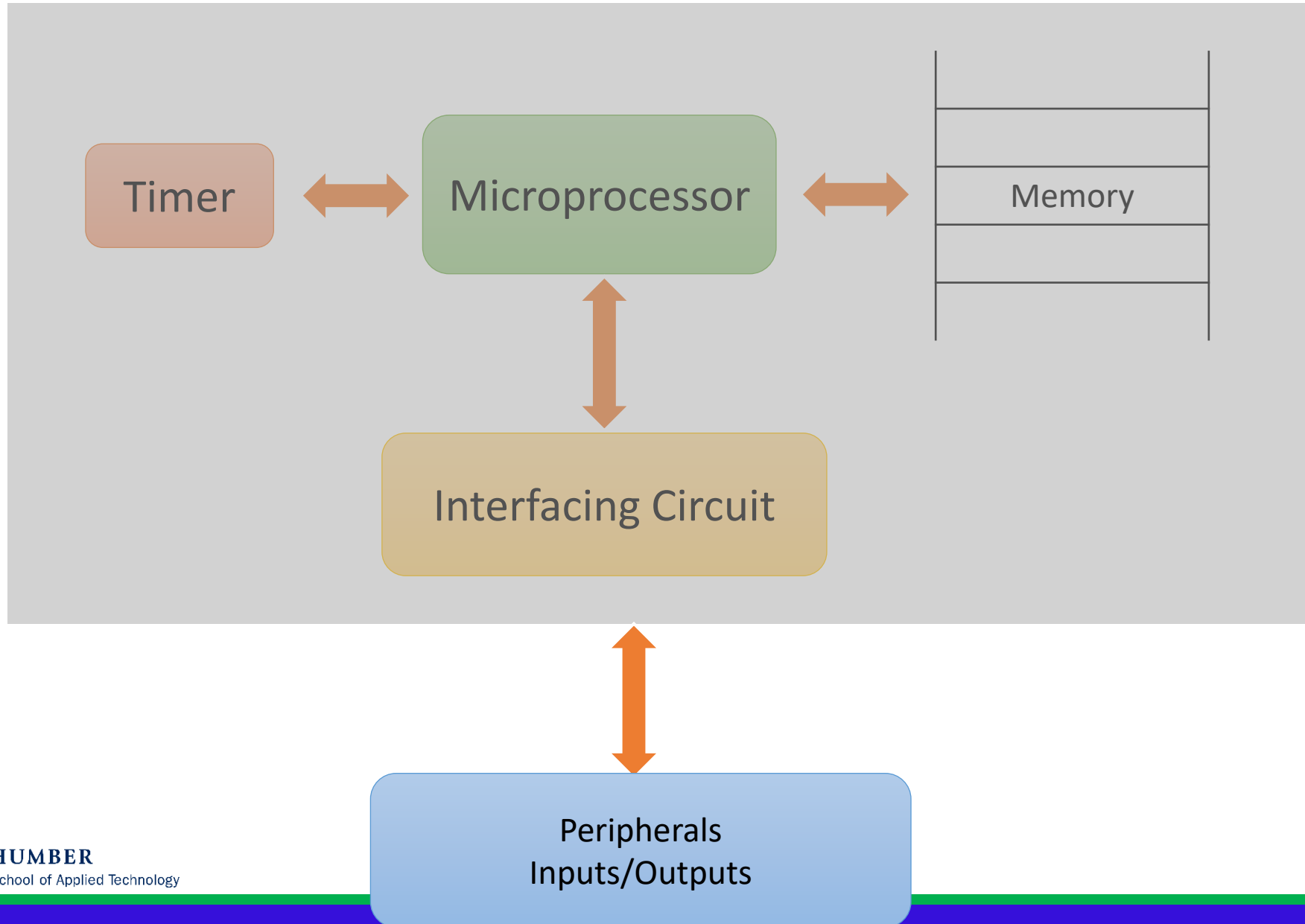
Core components:

- Microcontroller or Microprocessor.
- Memory: RAM, ROM, Flash.
- Sensors and Actuators.
- Communication Interfaces: UART, SPI, I2C.

Example



Example



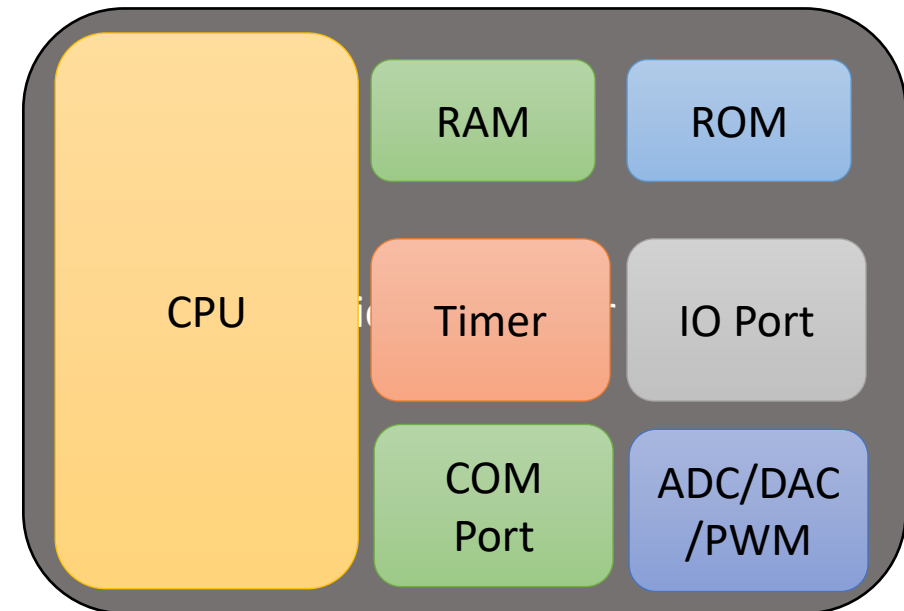
What is a microcontroller?

Definition: A compact integrated circuit that contains a CPU, memory, and peripherals.

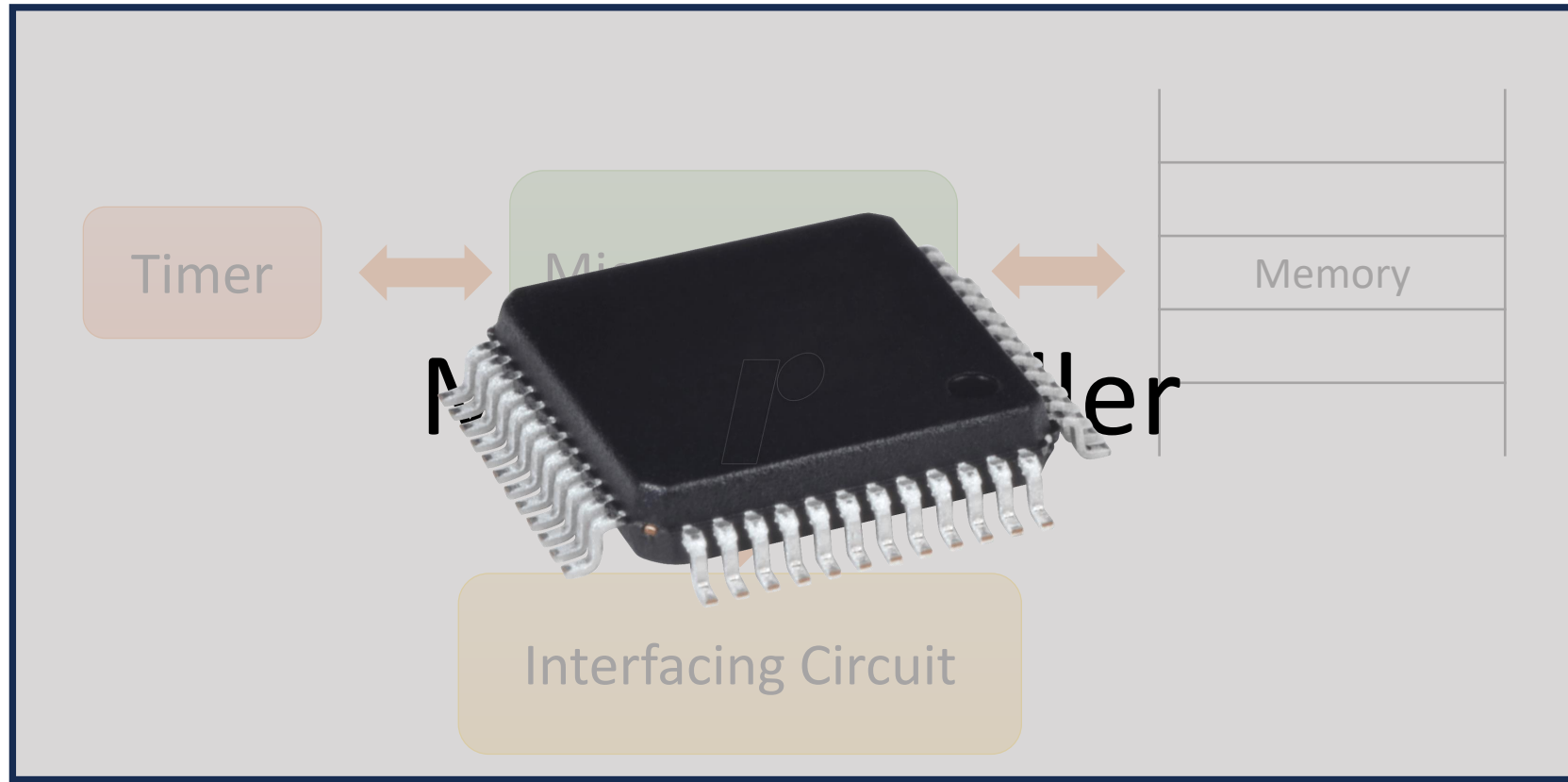


What is a microcontroller?

Definition: A compact integrated circuit that contains a CPU, memory, and peripherals.



Example



In-class Activity

- Pick one of the following embedded systems and indicate what are the main components:
 - Dishwasher
 - Microwave
 - Airfryer
 - Fridge with ice dispensing unit

Features of Microcontrollers

- Low power consumption.
- Integrated peripherals (timers, GPIO, ADC/DAC).
- Cost-effective for specific tasks.
- Variety of sizes and capabilities (8-bit, 16-bit, 32-bit).

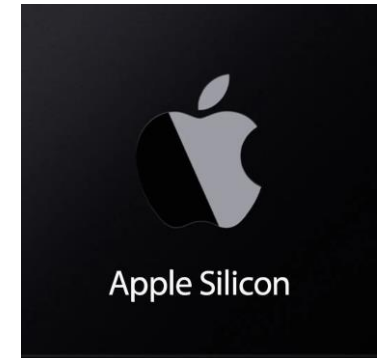
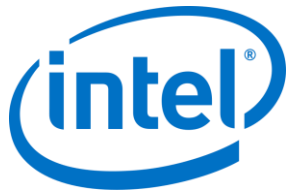
Big names in Microcontroller Industry

- Microchip Technology
- Texas Instrument
- STMicroelectronics
- NXP Semiconductors
- Renesas Electronics
- Infineon
- Espressif



What is the difference between microcontroller and microprocessor?

- Microcontrollers are for specific tasks, while microprocessors are for general-purpose tasks.



Microprocessor Limitations

- **No on-chip memory**- the designer needs to add external memory chips and other logic circuits such as decoder and buffer chips to provide program and data storage.
- **Can not drive input and output devices directly** due to the lack of power drive or voltage level incompatibility.
- **Does not have peripheral functions** such as timer, interrupt, I/O ports, analog-to-digital converter (ADC), communication interface and so on. These functions must be implemented with external chips.

Due to these limitations, the product designed with a microprocessor can not be very compact.

Case Study 1- Line Following Robot

System Components:

- **Microcontroller**
- **Sensors:** e.g. Infrared (IR) sensors to detect the line. (input devices)
- **Actuators:** DC motors to control wheel movement. (output devices)
- **Other Components:** Motor driver, power supply.

Role of the Microcontroller:

- Reads data from IR sensors to determine the robot's position relative to the line.
- Processes the data and executes an algorithm to adjust motor speeds for steering.
- Sends control signals to the motor driver.

Case Study 2- Robotic Arm for Assembly

Objective:

To create a robotic arm capable of assembling components in a manufacturing line.

• System Components:

- **Microcontroller**
- **Sensors:** Force sensors for precision handling.
- **Actuators:** Servo motors for joint movement.
- **Communication:** UART for real-time feedback from the control system.

• Role of the Microcontroller:

- Processes sensor data to detect component position and ensure proper alignment.
- Sends PWM signals to servo motors for precise joint movement.
- Maintains safety limits by monitoring force and position sensors.

Programming Microcontrollers

- **Languages:** C, C++ (sometimes Assembly for optimization).
- **Tools:** IDEs (e.g., Keil, Arduino IDE, STM32CubeIDE).
- **Process:**
 - Write the code.
 - Compile and upload to the microcontroller.
 - Test with peripherals.

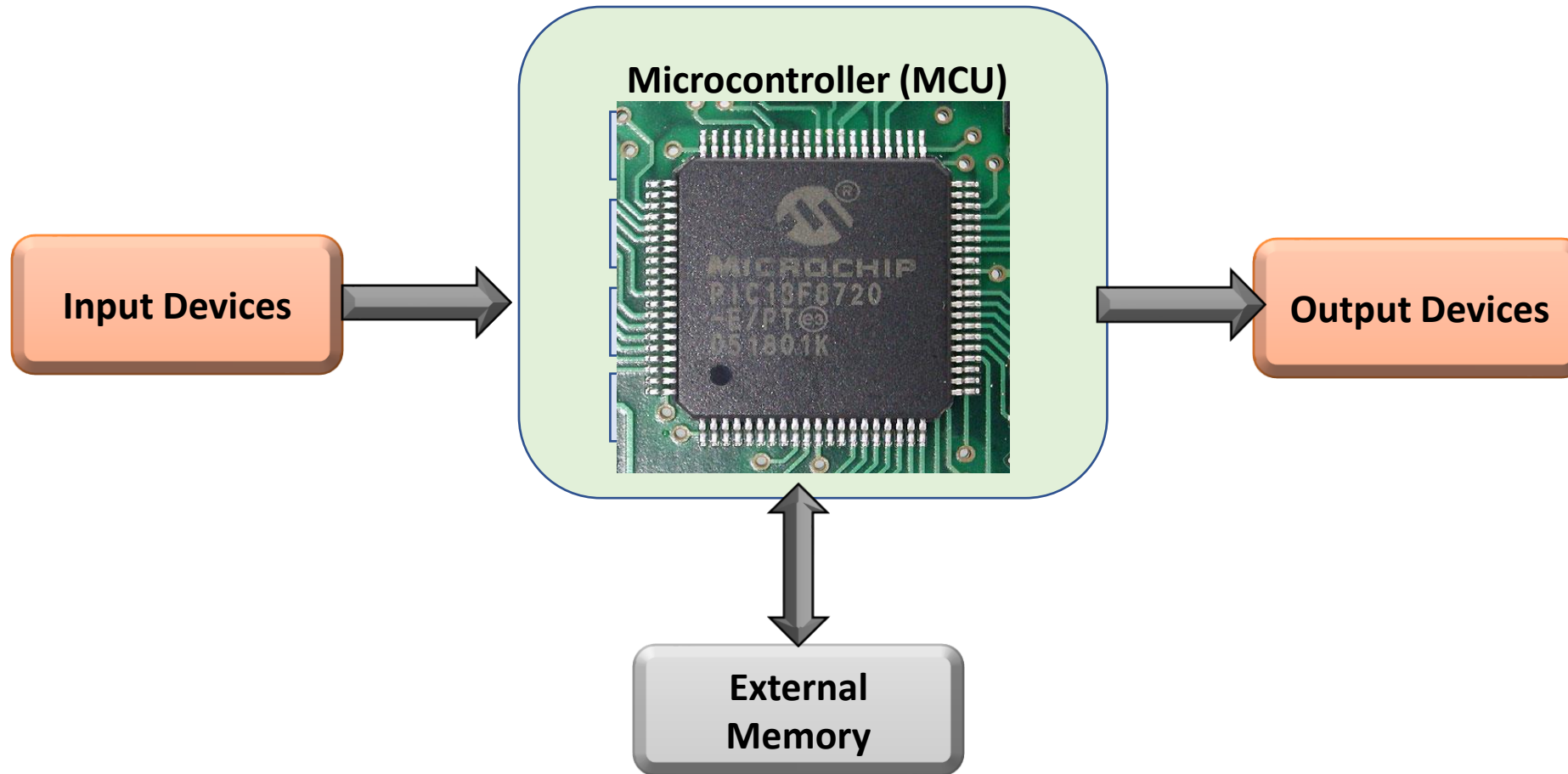
Role of Microcontrollers in Mechatronics

- **Integration of Mechanical and Electrical Systems:**
 - Microcontrollers serve as the "brain" of mechatronic systems.
 - Enable seamless interaction between sensors, actuators, and software.
- **Precision Control:**
 - Facilitate precise motor control, such as in robotic arms or drones.
 - Enable accurate readings from sensors to adjust operations dynamically.
- **Automation:**
 - Automate processes in manufacturing, such as assembly lines.
 - Control and monitor systems in real-time.
- **Connectivity:**
 - Enable communication between components using protocols like UART, SPI, I2C.
 - Allow integration with IoT systems for remote monitoring and control.
- **Energy Efficiency:**
 - Low-power microcontrollers enhance battery-operated mechatronic devices.

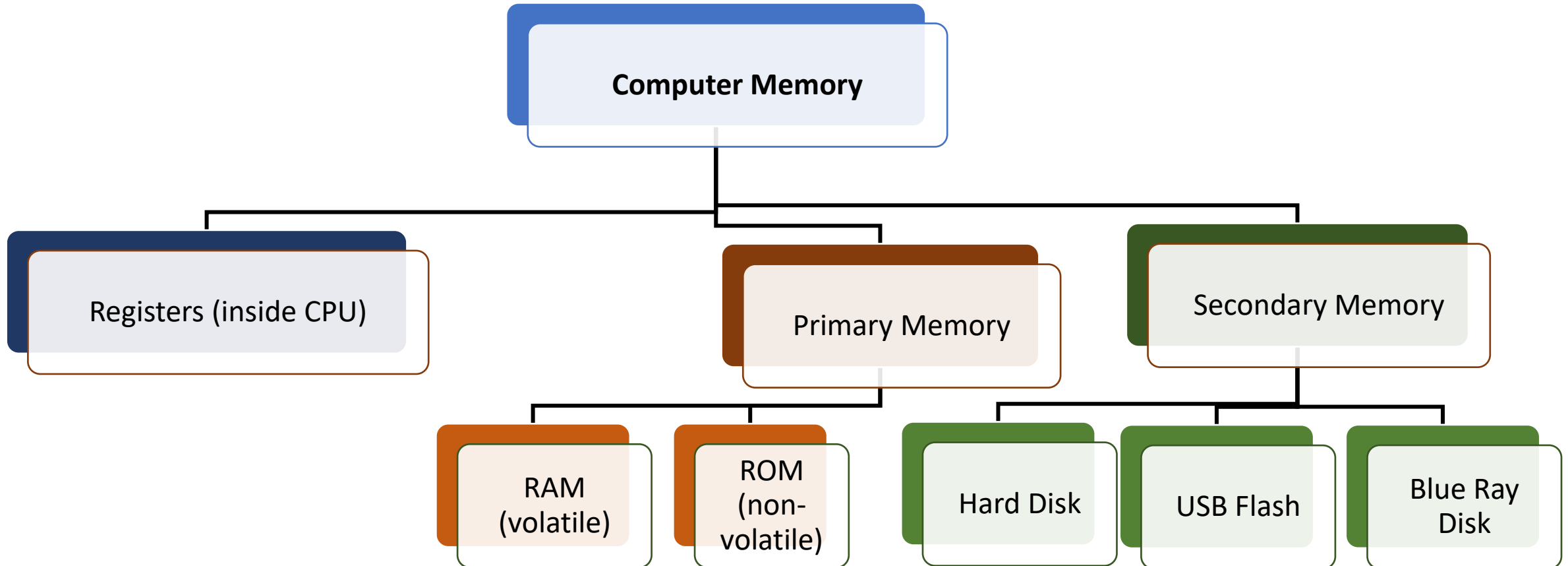
Future Trends in Embedded Systems for Mechatronics

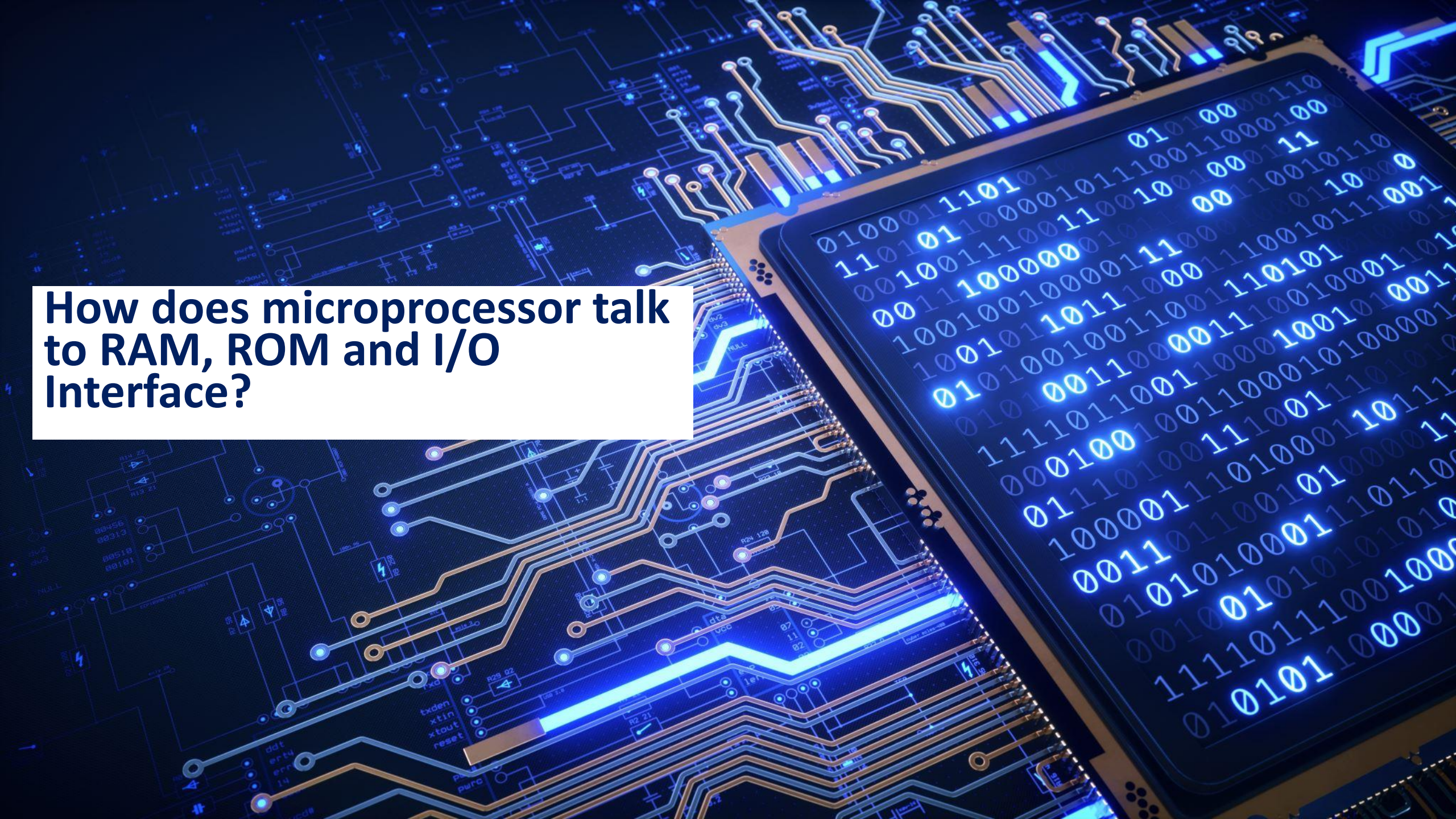
- Integration of AI and Machine Learning for decision-making.
- Use of edge computing for real-time processing.
- Enhanced IoT connectivity for smart manufacturing.

Building Blocks of a Typical Microcontroller



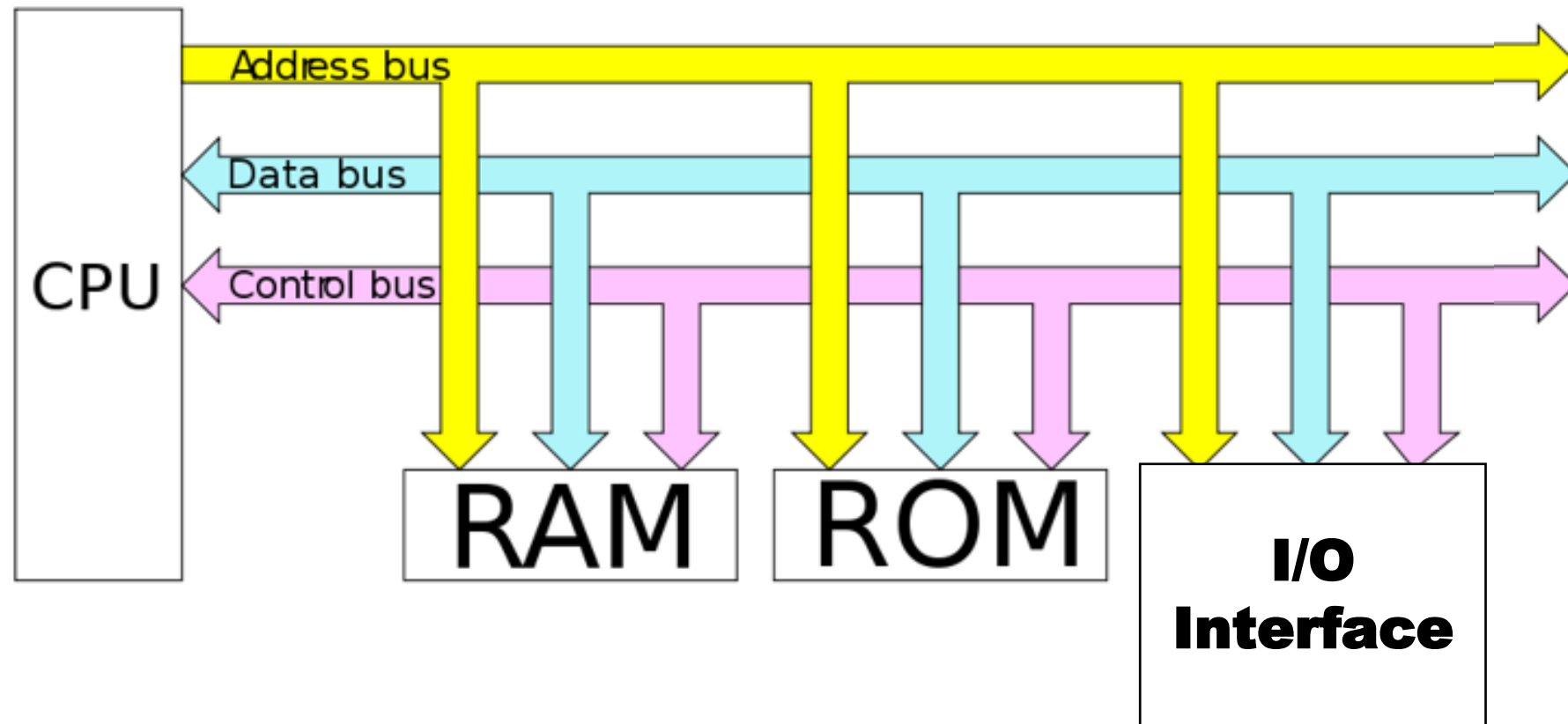
Memory Types



The background of the image is a detailed, high-tech circuit board. It features a complex network of glowing blue lines representing electrical traces and data paths. On the right side, there is a large, dark rectangular area filled with glowing blue binary code (0s and 1s) arranged in a grid-like pattern. The overall color scheme is dominated by deep blues and bright, neon-like blues, giving it a futuristic and digital feel. The circuit board itself has various components like capacitors, resistors, and integrated circuits visible, though they are less distinct due to the glowing effects.

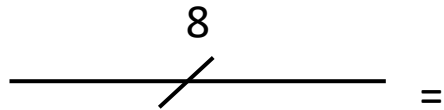
**How does microprocessor talk
to RAM, ROM and I/O
Interface?**

System Bus Architecture

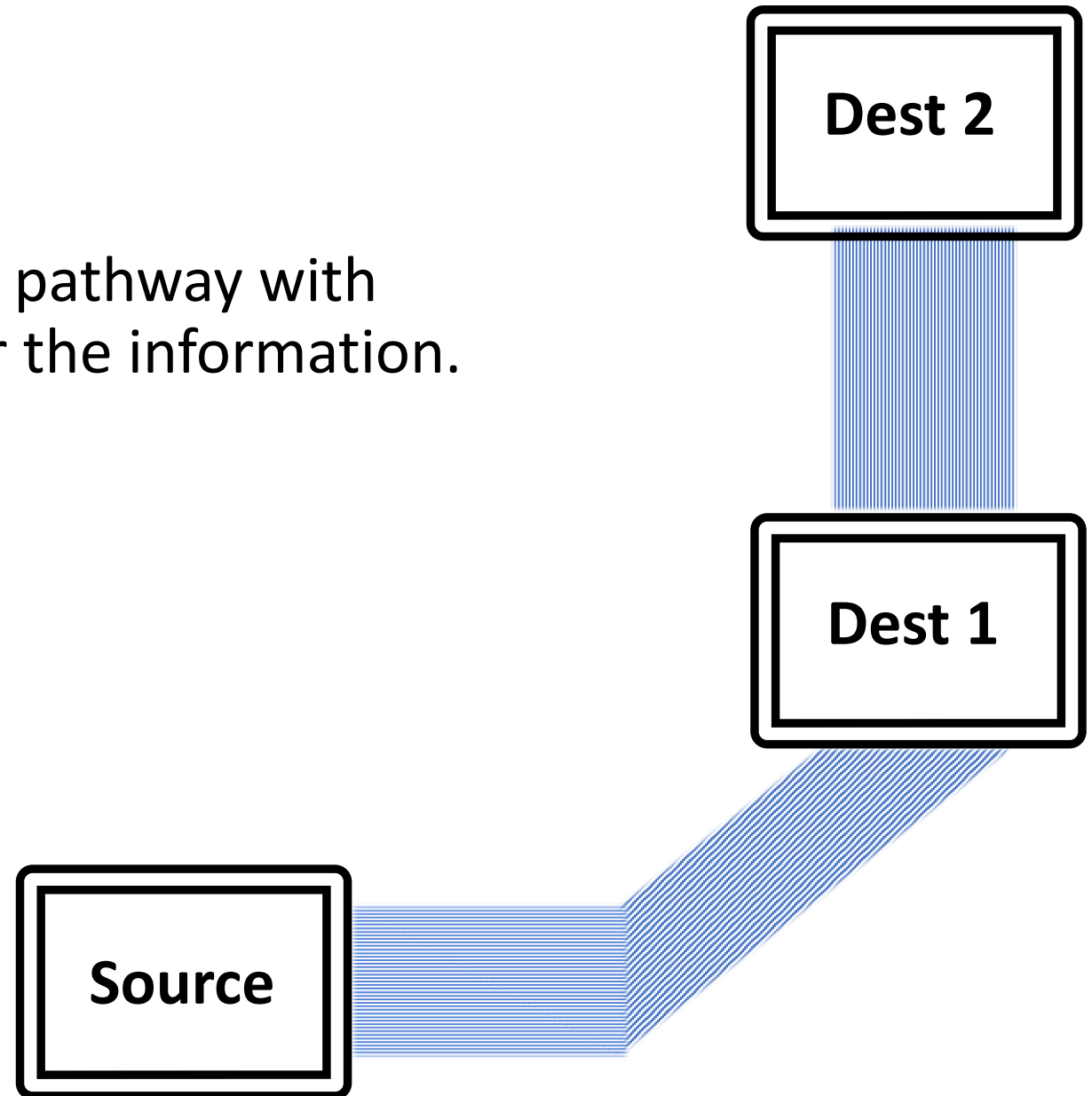


What is a BUS?

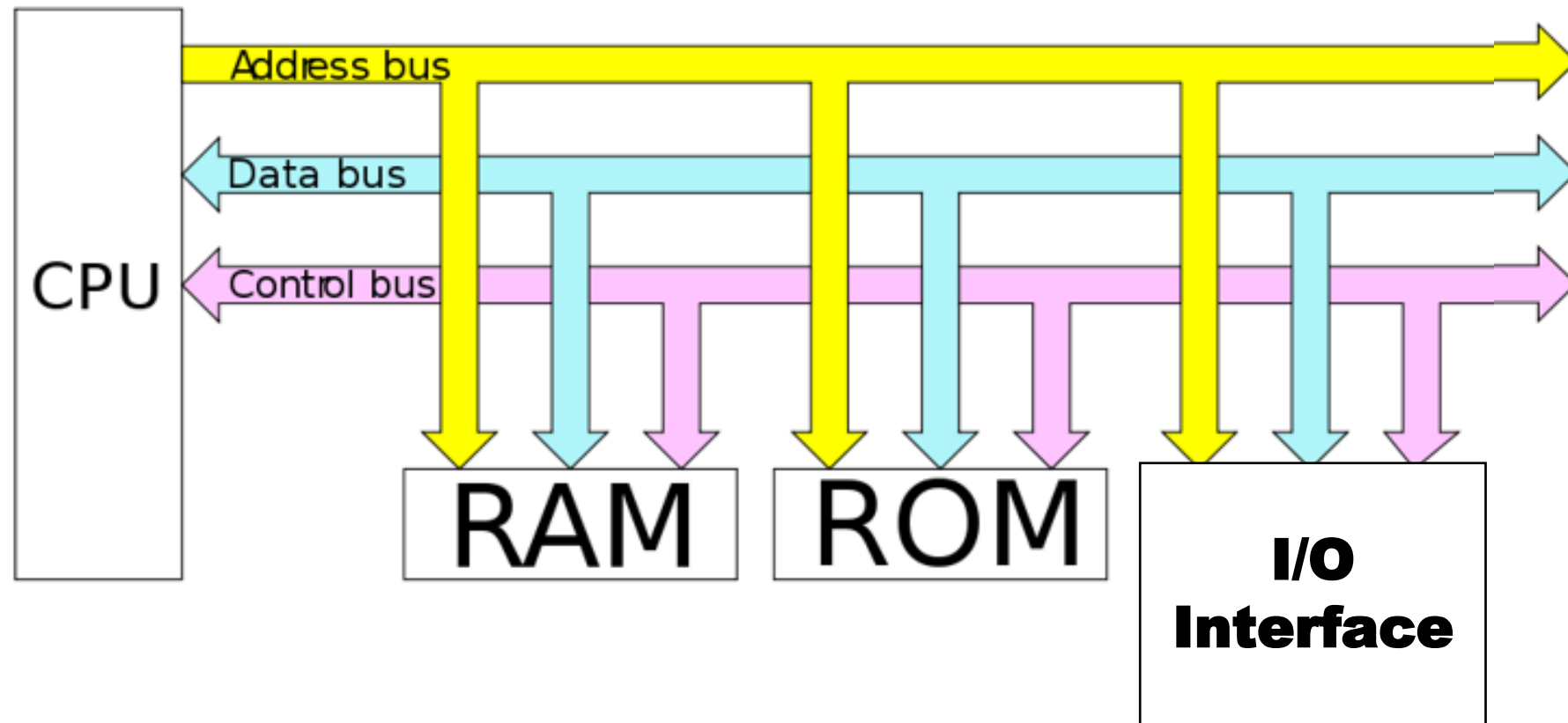
A bus is a multiple wire, information pathway with multiple sources and destinations for the information.



8-bit BUS

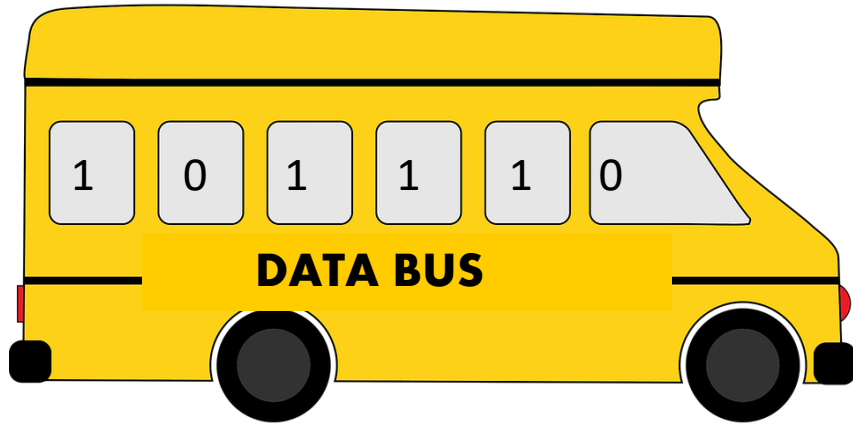


System BUS



What is a DATA BUS?

Size of data bus indicates the **largest packet of data** that can be transmitted simultaneously.



Common Data Buses:

8-bit

16-bit

32-bit

64-bit



What is the impact of Data
Size on system performance?

What is an ADDRESS BUS?

CPU provides the address of the destination (i.e. particular location in memory map) on the Address bus.





What is the impact of size of address bus on the **Performance** of a microcontroller?



Does the size of address bus
have any impact on the
operating speed of
microcontroller?

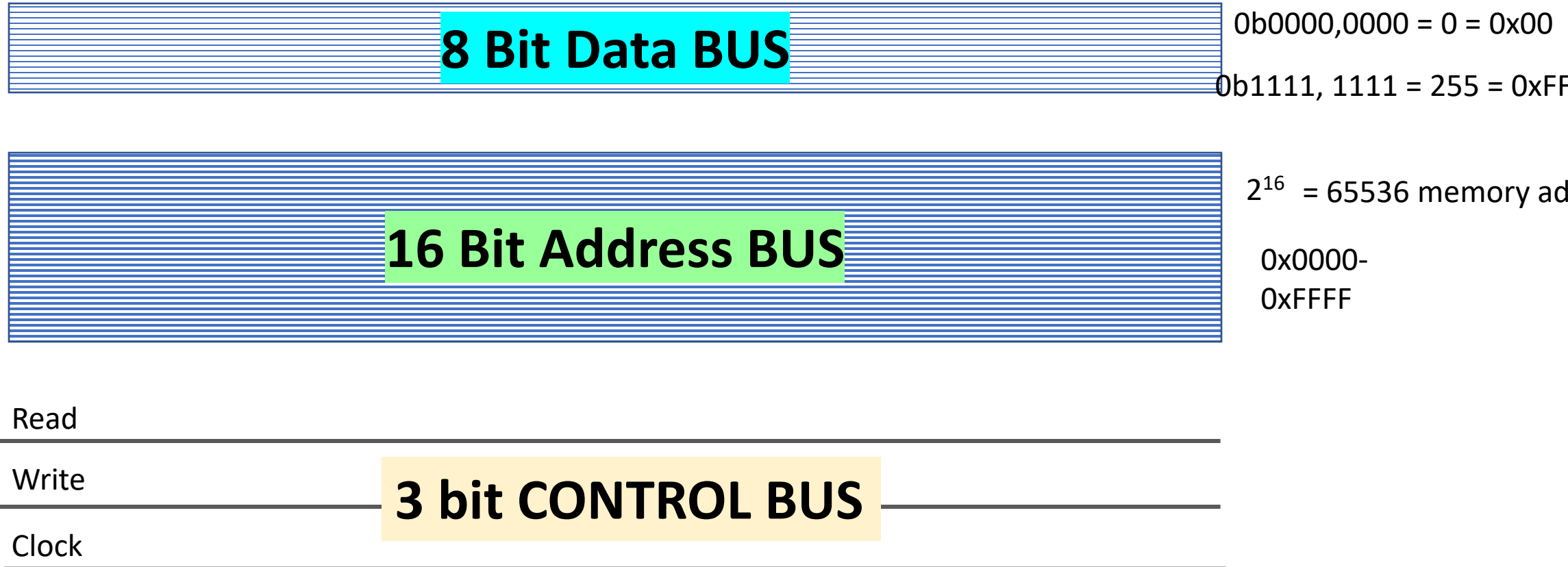
What is a Control BUS?



What is a Control BUS?



Simple BUS Example



Example of a Control Bus



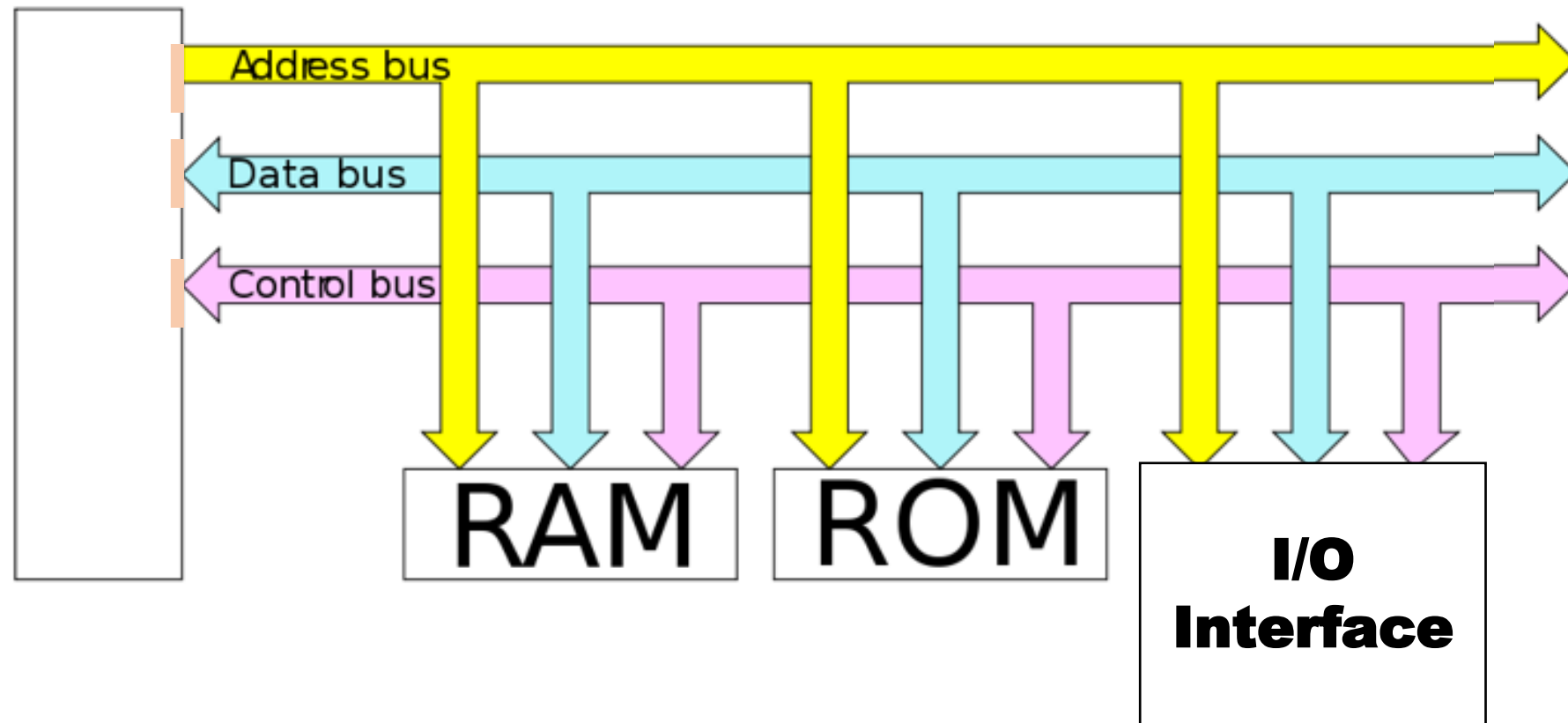
Read signal indicates that Data bus should read data from a particular memory location.

Write Signal indicates that Data bus should write data to a particular memory location.

Clock indicates the operating speed within the microcontroller system.

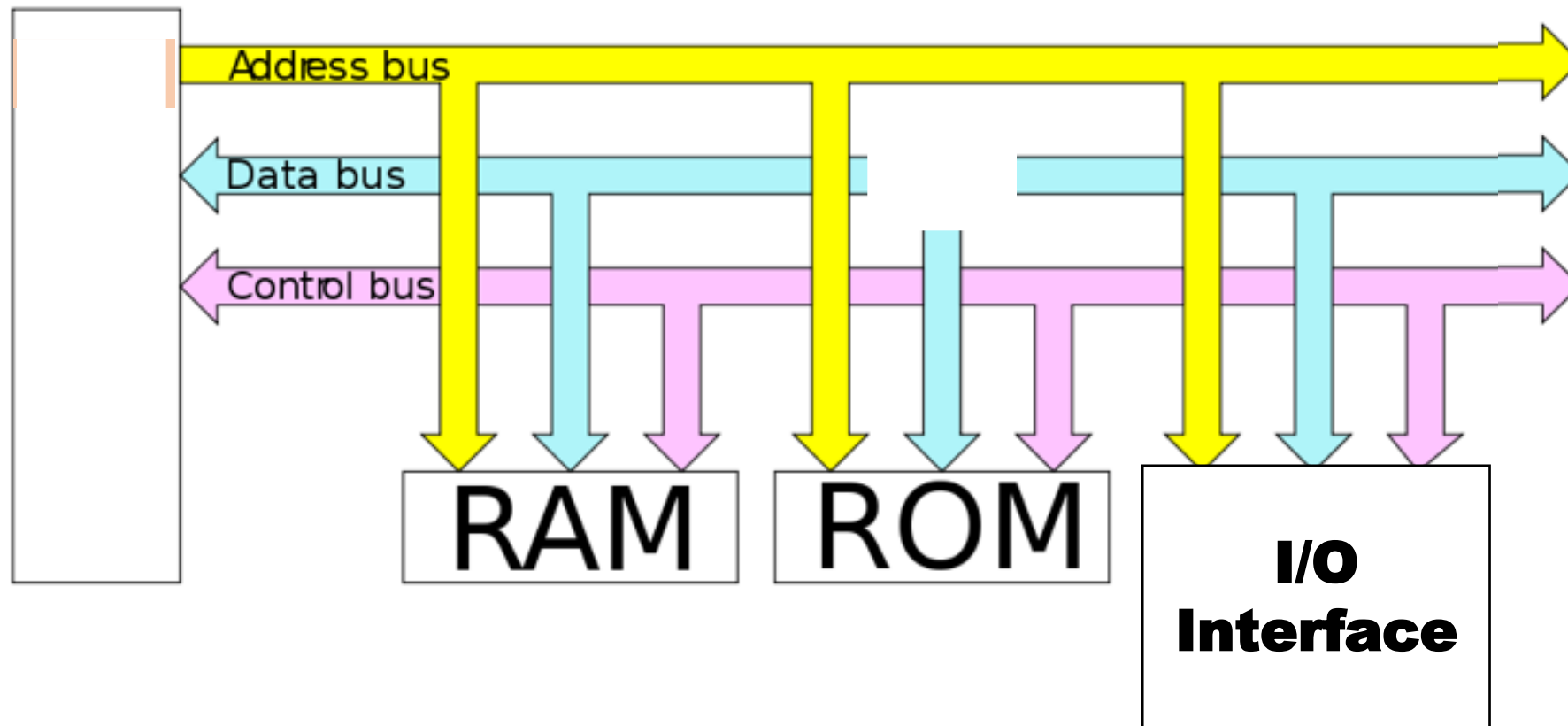
Example

Write “A= 0x41” to RAM at address 0x1014.



Example

Read the 3rd memory location of ROM



Conclusion

- Embedded Systems
- Microcontrollers
- Microcontrollers vs. Microprocessors
- Microcontrollers in Mechatronics
- Bus Interface for Microprocessor